

Yangtao Ge

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EDUCATION

- **University of Oxford** Oxford, UK
Master (Msc) in Software and Systems Security, Part-Time Sept. 2024 – Present
- **University College London** London, UK
Master of Engineering (MEng) in Computer Science; Degree Award: First Class Honours Sept. 2018 – Sept. 2022
- **University College London** London, UK
Undergraduate Preparatory Certificate; Avg. Score 86/100 Sept. 2017 – June. 2018

PROFESSIONAL SUMMARY

Senior Backend Software Engineer with demonstrated expertise in cloud architecture optimization, driving 70% latency reduction and 50% cost savings at Smartsheet. Architect of scalable microservices and event-driven systems using Golang, Java, and Python with AWS (EKS/ECS). Proven leader who resolves critical enterprise escalations while mentoring junior engineers. Combines technical excellence in CI/CD automation and infrastructure-as-code with cross-functional collaboration skills, consistently delivering measurable performance improvements that translate to business value.

PERSONAL SKILLS

- **Programming Languages:**
 - **Python:** Proficient in building distributed asynchronous systems data pipelines. Experienced with Apache Arrow for efficient Parquet file operations, multiprocessing for parallel data transformations
 - **Golang:** Proficient in building concurrent applications, primarily for microservices using Chi and Gin frameworks.
 - **Java:** Proficient in building scalable server-side applications using Spring Framework and Gradle. Experienced with Apache Flink for stream processing and real-time data ingestion from AWS Kinesis.
 - **Other Languages:** C/C++ (Competent); JavaScript (Proficient); SQL (Proficient)
- **Cloud Platform & Tools:**
 - **AWS:** Experienced with EC2, SQS, S3, and Lambda for cloud-based applications building with **terraform**
 - **Docker:** Strong capabilities in containerization and version control of deployment environments.
 - **Kubernetes & ECS:** Proficient in orchestrating and managing containerized applications in AWS EKS and ECS
 - **Apache Flink & Kinesis:** Experienced in building real-time stream processing pipelines with Apache Flink and AWS Kinesis for high-throughput data ingestion
- **DevOp & Analytic:**
 - **GitLab/GitHub:** Proficient with git for version control and building CI/CD pipelines with GitLab Pipeline.
 - **OpenSearch:** Implemented for search and log analysis solutions using Fluentd in containerized environments.
 - **Datadog & Grafana:** Proficient in building interactive dashboards for real-time data visualization and analytics.

WORK EXPERIENCE

- **Smartsheet** London, UK
Software Engineer II Jan. 2023 - Present
 - **Technical Innovation & Leadership:**
 - * Architected and led the Async Move History Project using Python, building a distributed asynchronous system with multiple workers orchestrated through AWS SQS and DynamoDB distributed lock tables. Leveraged Apache Arrow for efficient Parquet file read/write operations with concurrent data transformations while maintaining optimal memory usage, eliminating API gateway timeout errors and reducing system latency by 70%.
 - * Designed Python-based distributed processing pipeline with SQS message queuing and DynamoDB for coordination, orchestrating parallel workers for data transformation tasks while ensuring data integrity and efficiency through Apache Arrow's columnar processing capabilities.

- * Spearheaded cross-functional collaboration with other service teams to resolve two critical customer escalations affecting enterprise clients with "whale grids" (10M+ history records).
 - * Achieved 50% cost reduction through strategic system redesign, legacy system deprecation, and optimization of API service resources, delivering significant ROI.
 - **DevOps Excellence:**
 - * Designed and implemented a company-wide centralized ECS pipeline (adopted as part of PEAK initiative) that standardized deployment processes and improved security posture, also slashed the new service onboarding time from weeks to under 24 hours through template standardization and process automation.
 - * Pioneered the adoption of Graviton instances and multi-architecture builds, reducing ECS operational costs by 25% while maintaining performance.
 - * Created hardened, secure container images with automated vulnerability scanning, establishing new security standards for the organization.
 - **Core Technical Contributions:**
 - * Developed and optimized Golang microservices (CHI framework) that improved P95 latency by 20% through efficient goroutine parallelization and caching strategies.
 - * Built real-time customer actions ingestion pipeline using Apache Flink to process high-volume AWS Kinesis streams, implementing parallel data paths: writing to S3 as Parquet files for cost-effective cold storage and simultaneously ingesting to DynamoDB for low-latency hot data access, handling millions of events per second with sub-second latency.
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- **Spotify** London, UK
Backend Engineer Intern *June. 2022 - Sept. 2022*
 - **Java Microservices:** Developed backend microservices using the Apollo framework and gRPC, which enhances music delivery processes and service efficiency for both music providers and Spotify operatives.
 - **Testing Enhancement:** Authored technical documentation outlining key strategies for testing improvements, which was instrumental in increasing code coverage from 75% to 90%, substantially boosting application reliability
 - **Operational Integration:** Engineered a Slack notification workflow that automatically informs Spotify operatives of stakeholder status updates, enhancing operational communication and responsiveness.
 - **R3** London, UK
Solution Engineer *June. 2021 - Sept. 2021*
 - **Confidential Computing Development:** Actively contributed to the enhancement of the Conclave example codebase utilizing Kotlin and Java, which has been utilized in commercial sales processes, communicating with non-technical stakeholders to demonstrate practical applications of confidential computing in business solutions
 - **Corda Certification:** Successfully passed Corda Developer Certification Exam, validating expertise in enterprise blockchain technology such as distributed ledger and contracts. [Credly Credential]

PROJECTS

- **Fuzz Testing for DPC++** London, UK
Individual Project Researcher *Sept. 2021 - Jun. 2022*
 - **DPC++ Fuzzing Framework:** Wrote a dissertation to propose the first fuzzing framework for DPC++ which includes an instrument tool to track the coverage and a fuzzer to generate test inputs. Establish a DPC++ code collection to benchmark the testing framework effectiveness.
 - **LLVM & Clang:** Implement the testing framework under LLVM compiler infrastructure and use Clang LibTooling API to instrument target program automatically on AST level.
- **Ophidia Technologies: Smart Label Project** London, UK
Tech Team Leader & Full stack developer *Jun. 2021 - Dec. 2021*
 - **Leadership and Architecture Design:** Led a team of three software engineers in the design and development of a full stack application for smart e-ink labels. Spearheaded the software architecture planning, ensuring optimal integration of front-end and back-end components for enhanced performance and scalability.
 - **Dockerization and Pipeline:** Successfully dockerized the backend API, facilitating smoother, more reliable deployments. Engineered and deployed a CI/CD pipeline on AWS, improving development workflow and deployment efficiency by automating updates and testing processes.